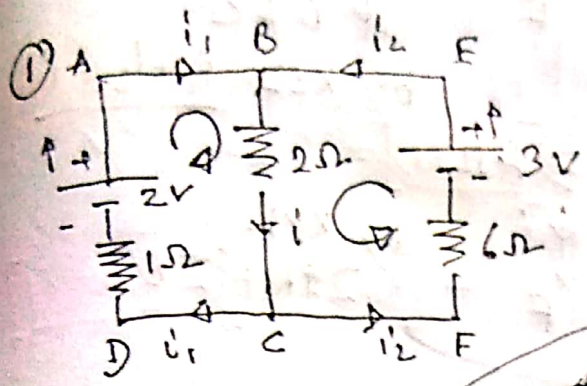
Solution of Problems on Kirchoff's LaWs

(1)



By KCL $i = i_1 + i_2$ at B & C.

By KVL

$\Sigma IR = \Sigma E$

In loop ABCDA, $2i + 1 \times i_1 = 2$

$2(i_1 + i_2) + i_1 = 2$

$\boxed{3i_1 + 2i_2 = 2}$ — (1)

In loop EBCFE, $2i + 6i_2 = 3$

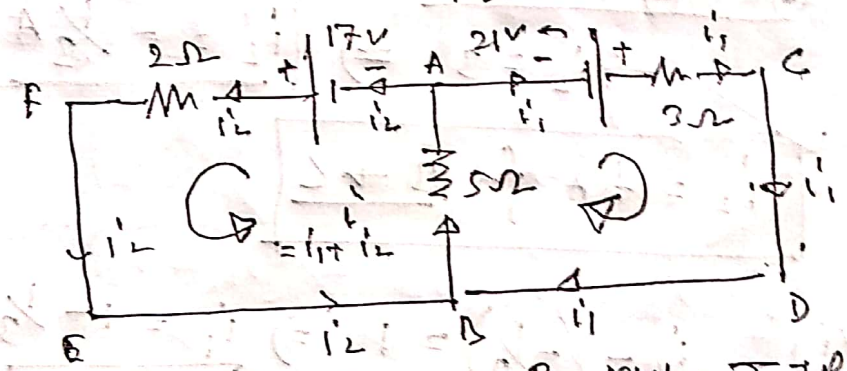
$2(i_1 + i_2) + 6i_2 = 3$

$\boxed{2i_1 + 8i_2 = 3}$ — (2)

Solving (1) & (2) we get $\boxed{i_1 = \frac{1}{2} A = 0.5A}$

& $\boxed{i_2 = \frac{1}{4} A = 0.25A}$

(2)



at B, A $i = i_1 + i_2$ (KCL)

By KVL $\Sigma IR = \Sigma E$

in loop ACDBA, $3i_1 + 5i = 21$

$3i_1 + 5(i_1 + i_2) = 21$

$\boxed{8i_1 + 5i_2 = 21}$ — (1)

In loop AFEBA, $2i_2 + 5i = 17$

$2i_2 + 5(i_1 + i_2) = 17$

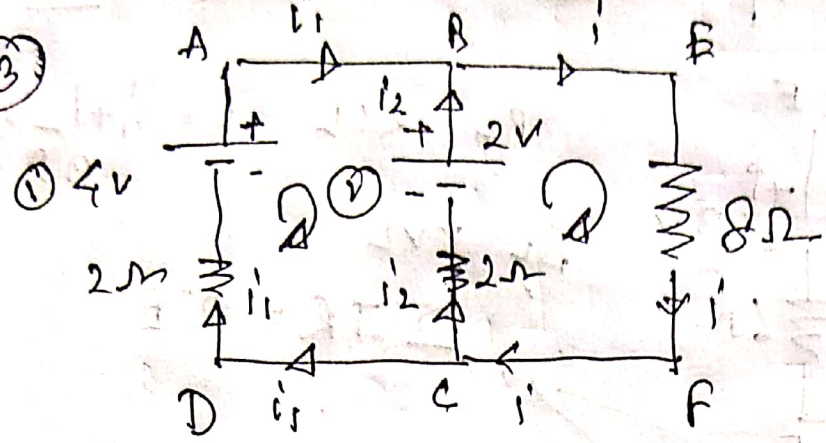
$\boxed{5i_1 + 7i_2 = 17}$ — (2)

Solving (1) & (2) we get $i_2 = 1A$, $i_1 = 2A$

& $i = i_1 + i_2 = 3A$

Scanned with CamScanner

③



at B & C

by KCL $i_1 + i_2 = i$

By KVL

$\Sigma IR = \Sigma E$

in loop ABCDA,

$$-2i_2 + 2i_1 = -2 + 4$$

$$\boxed{i_1 - i_2 = 1} \quad \text{--(i)}$$

in loop BEFCB

$$8i + 2i_2 = 2$$

$$4(i_1 + i_2) + i_2 = 1$$

$$\boxed{4i_1 + 5i_2 = 1} \quad \text{--(ii)}$$

(i) × 4 → $4i_1 - 4i_2 = 4$

(ii) → $(-)\,4i_1 \;(-)\; 5i_2 = (-)\,1$

$$-9i_2 = 3 \Rightarrow \boxed{i_2 = -\tfrac{1}{3}\,A}$$

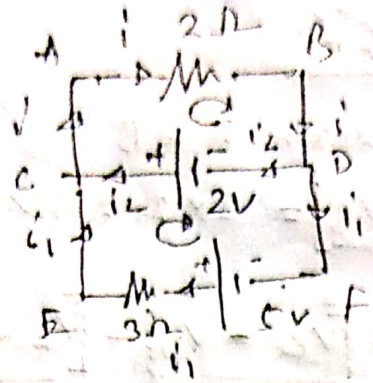
$$\boxed{i_2 = \tfrac{1}{3}\,A \text{ from } B \to C}$$

from (i) $i_1 - (-\tfrac{1}{3}) = 1 \Rightarrow i_1 = 1 - \tfrac{1}{3}$

$$\boxed{i_1 = \tfrac{2}{3}\,A}$$

Scanned with CamScanner

(4)



at C & D

KCL $i = i_1 + i_2$.

By KVL $\Sigma IR = \Sigma E$

in loop ECDFE

$3i_1 = 5 - 2$

$i_1 = 3/3 = 1A$

$\boxed{i_1 = 1A}$

in loop DCABD,

$2i = 2$

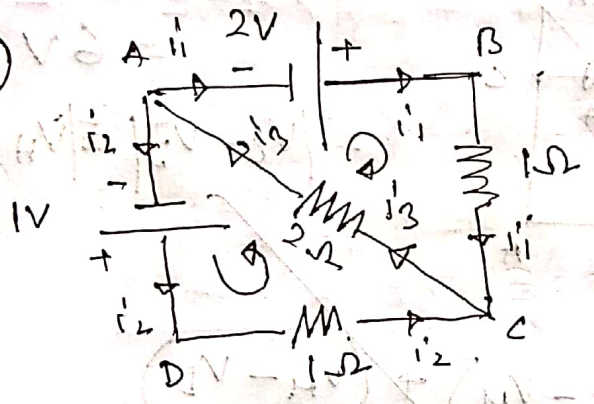
$i = 1A$

$\therefore$ PD acrm AB i.e 2Ω $V_{AB} = i \times 2$

$= 2 \times 1$

$= 2V$

(5)



at C & A by KCL $i_3 = i_1 + i_2$

by KVL, $\Sigma IR = \Sigma E$

in loop ABCA,

$1 \times i_1 + i_3 \times 2 = 2$

$i_1 + (i_1 + i_2)2 = 2$

$\boxed{3i_1 + 2i_2 = 2}$ — (1)

& in loop ADCA, $1 \times i_2 + i_3 \times 2 = 1$

$i_2 + (i_1 + i_2) \times 2 = 1$

$\boxed{2i_1 + 3i_2 = 1}$ — (2)

Solving (1) & (2)

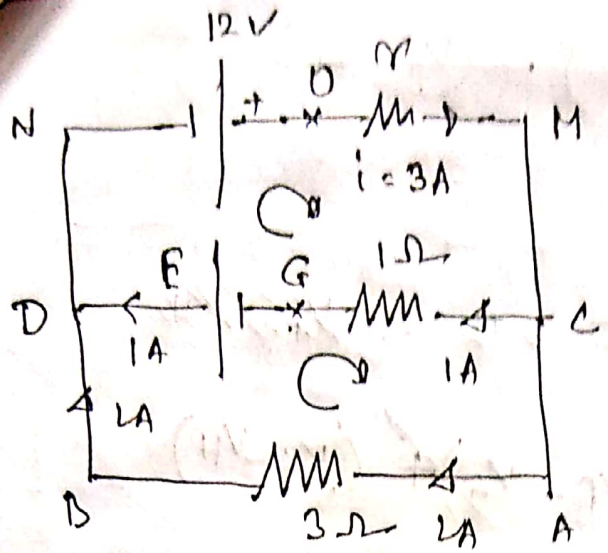
we get $\boxed{i_2 = -\frac{1}{5} = -0.2A}$

& $\boxed{i_1 = \frac{8}{10} = 0.8A}$

$\therefore i_3 = i_1 + i_2 = 0.8 - 0.2$

$\boxed{i_3 = 0.6A}$

Scanned with CamScanner



Method-1 at C & D by KCL $i = 1+2 = 3A$

By KVL $\Sigma IR = \Sigma E$

in loop ABDCA, $2\times3 - 1\times1 = -E$

$$\boxed{E = -5V}$$

in loop CDNMC, $1\times1 + 3\times r = E + 12$

$$1 + 3r = -5 + 12$$

$$3r = 7 - 1$$

$$\boxed{r = 2\Omega}$$

Method 2

$$V_A - V_B = 2\times3 = 6V$$

$$= [V_C - V_D] = [V_M - V_N]$$

$$V_C - V_D = (V_C - V_G) + (V_G - V_D)$$

$$6 = 1\times1 - \underbrace{(V_D - V_G)}_{E}$$

$$6 = 1 - E \Longrightarrow \boxed{E = 5V}$$

Scanned with CamScanner

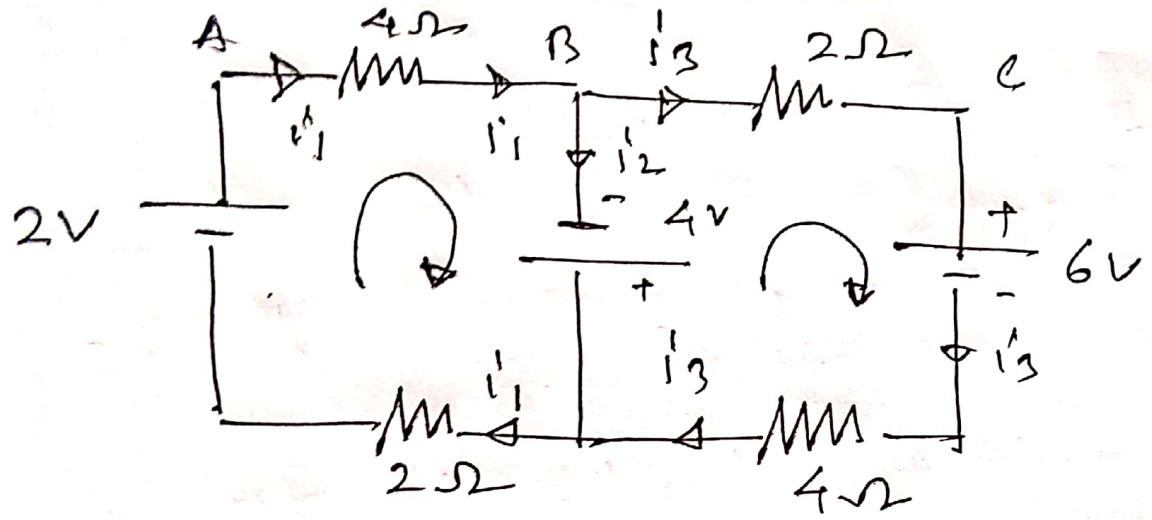
$$V_M - V_N = (V_M - V_O) + (V_O - V_N)$$

$$6 = -(V_O - V_M) + (V_O - V_N)$$

$$6 = -3r + 12$$

$$3r = 6 \Rightarrow \boxed{r = 2\Omega}$$

(7)



By KCL at B $\quad i_1 = i_2 + i_3$

By KVL $\quad \Sigma IR = \Sigma E$

in Loop ABEFA,

$$4i_1 + 2i_1 = 4 + 2$$

$$6i_1 = 6 \quad \therefore \boxed{i_1 = 1A}$$

in Loop BCDEB,

$$2i_3 + 4i_3 = -6 - 4$$

$$6i_3 = -10, \quad \boxed{i_3 = -\frac{5}{3}A}$$

$$\therefore i_1 = i_2 + i_3 \Rightarrow i_2 = i_1 - i_3 = 1 + \frac{5}{3}$$

$$\boxed{i_2 = \frac{8}{3}A}$$

Scanned with CamScanner